

# Phonemic Analysis

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```
source(here::here("_defaults.R"))
```

```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ lubridate 1.9.5     ✓ tibble    3.3.1
✓ purrr     1.2.2     ✓ tidyr     1.3.2
```

```
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(tinytable)
library(rlang)
```

```

Attaching package: 'rlang'

The following objects are masked from 'package:purrr':

    flatten, flatten_chr, flatten_dbl, flatten_int, flatten_lgl,
    flatten_raw, invoke, splice
```



```
library(patchwork)
```

## Recap: Phonetics vs Phonology

**Phonetics**

| | |
|---|---|
| “pin” | [pʰɪn] |
| “spin” | [spɪn] |
| “bin” | [pɪn] |

One has a long-lag VOT and the other two have short-lag VOT

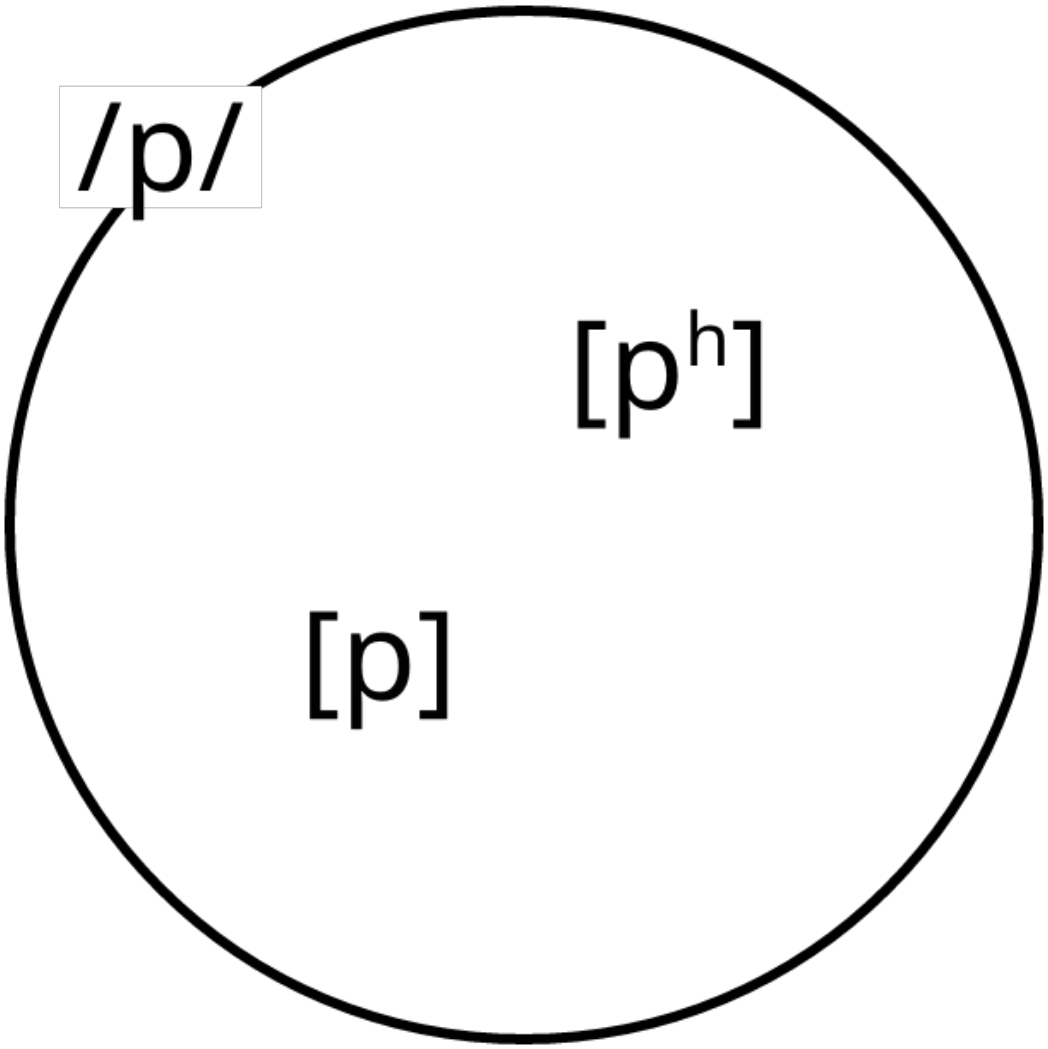
**Phonology**

| | |
|---|---|
| “pin” | /pɪn/ |
| “spin” | /spɪn/ |
| “bin” | /bɪn/ |

The first two are allophones of /p/, and the third is an allophone of /b/

## Recap: Phoneme

> Phoneme: A label for a set of sounds that all count as basically the same.




The phoneme /p/ has at least two *allophones*, [p] and [pʰ]

## Recap: Diagnostics

### Minimal Pairs

If there are at least two words that are differentiated by just these two sounds, they are *contrastive*.

### English

[d]

| | |
|---|---|
| “dough” | doʊ |
| “den” | dɛn |

[ð]

| | |
|---|---|
| “though” | ðoʊ |
| “then” | ðɛn |

## Recap: Diagnostics

### Complementary Distribution

If two sounds never appear in the same position, they are probably *allophonic*

### Spanish

Word initial

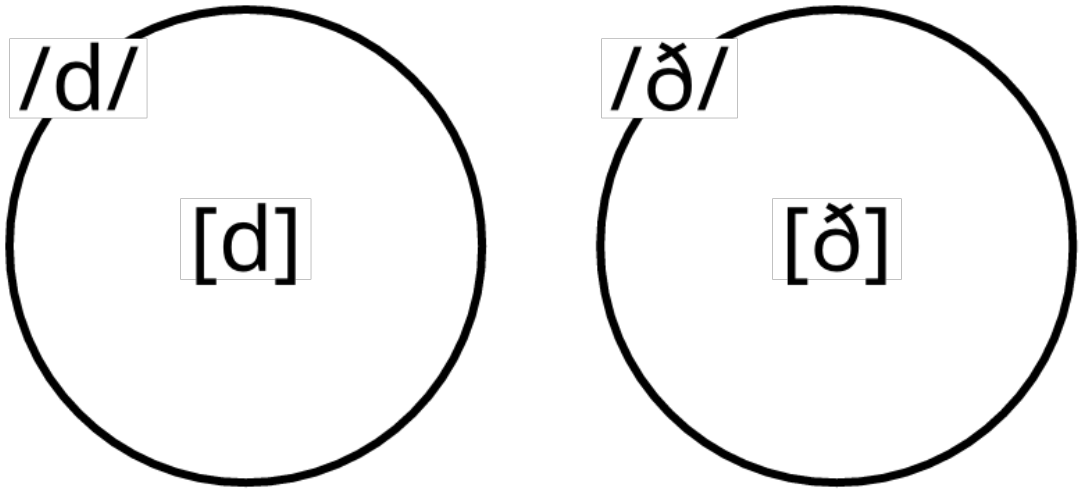
| | | |
|---|---|---|
| *lady* | “dama” | dɑmɑ |

Elsewhere

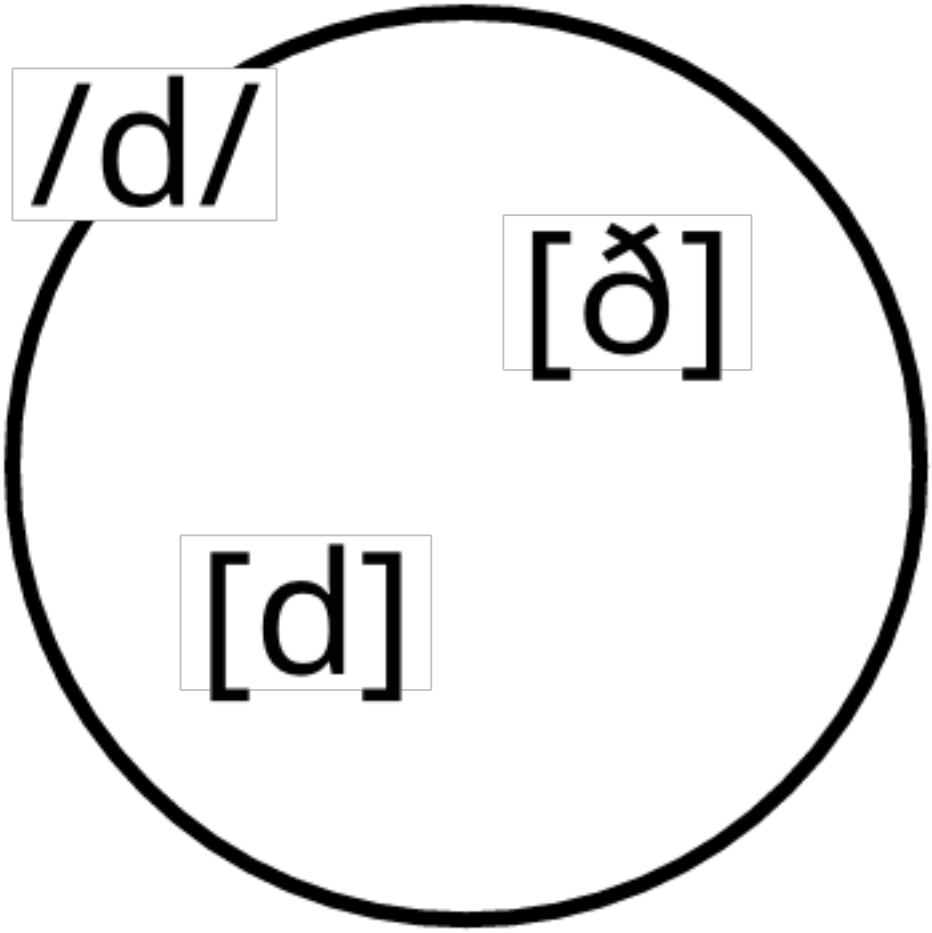
| | | |
|---|---|---|
| *side* | “lado” | lɑðo |

## Recap: Diagnostics

English:



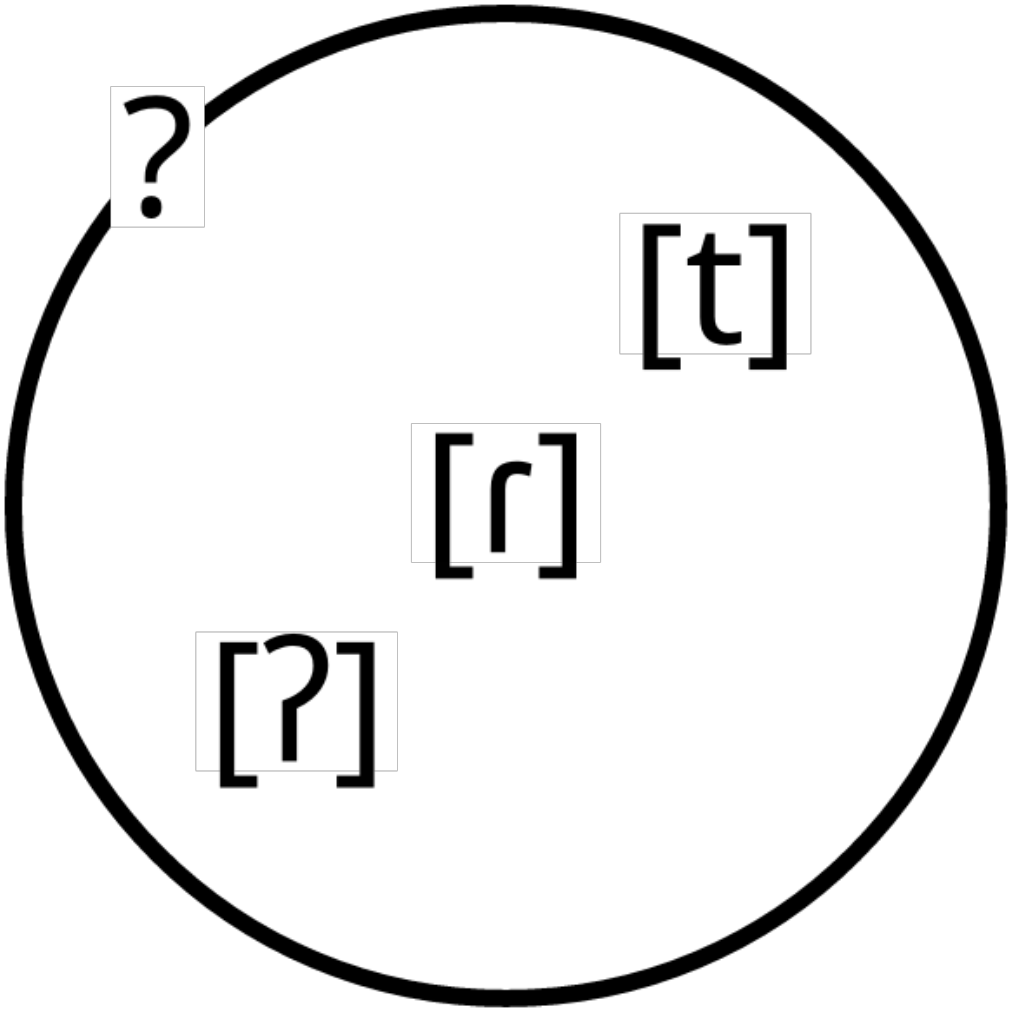
Spanish:




## Phonemic Analysis

What is the underlying form?

| | |
|---|---|
| *sit* | sɪt |
| *sitting* | sɪɾɪŋ |
| *sittin'* | sɪʔn̩ |

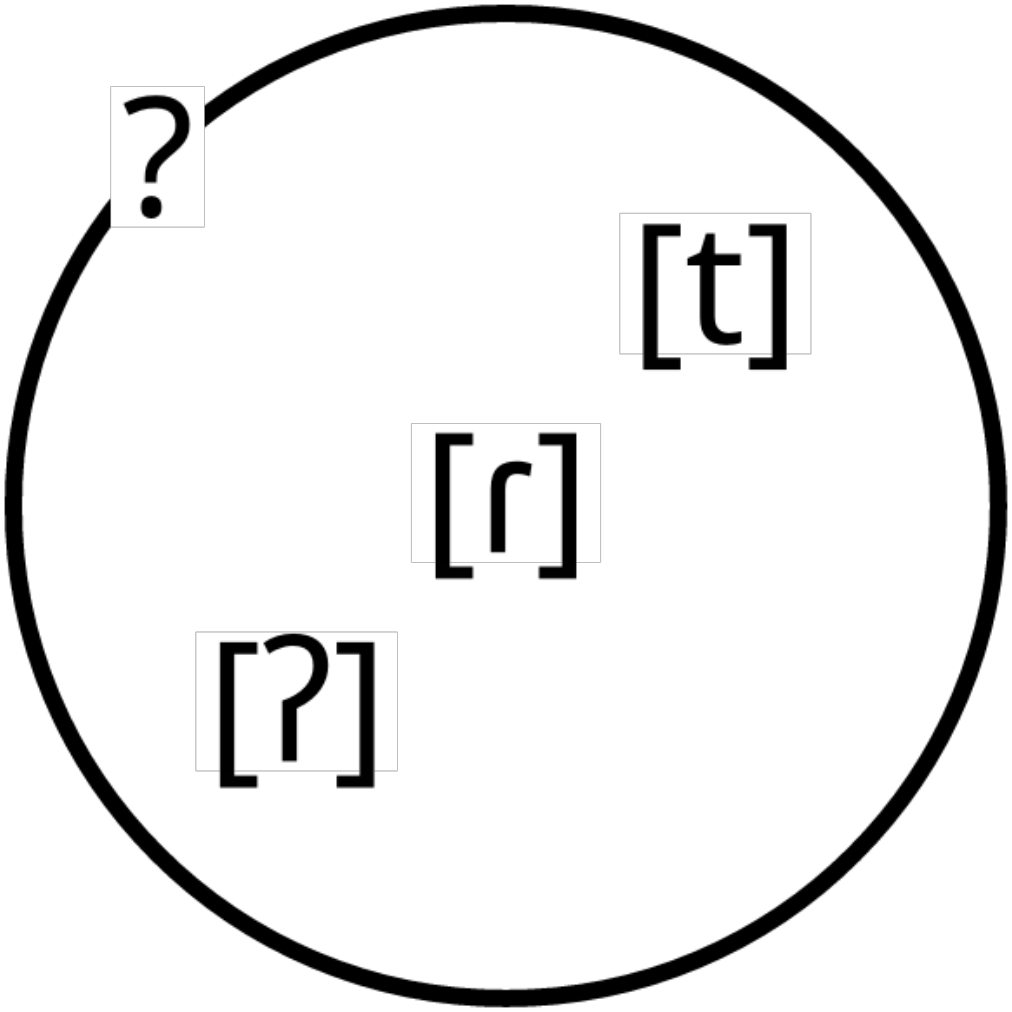



## Phonemic Analysis

What is the underlying form?

Choice of the underlying form here will indicate our thought about whether the phoneme is

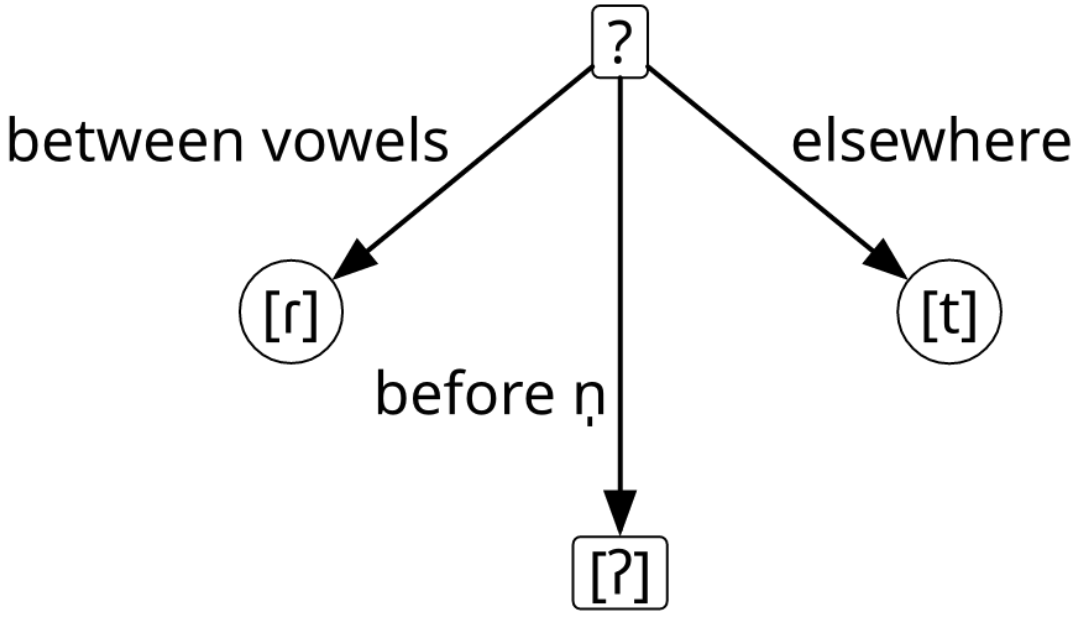
- Voiced or Voiceless
- A Stop or a Flap
- Coronal or Non-Coronal




## Phonemic Analysis

What is the underlying form?

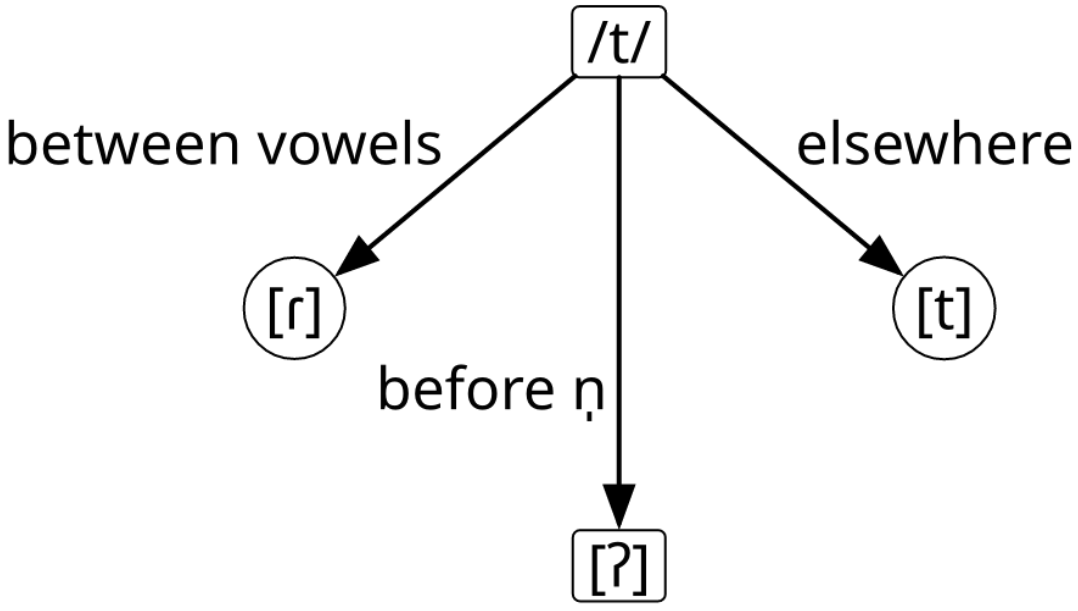
Which form shows up as the default?




## Phonemic Analysis

What is the underlying form?

Which form shows up as the default?





## Phonemic Analysis

What is the underlying form?

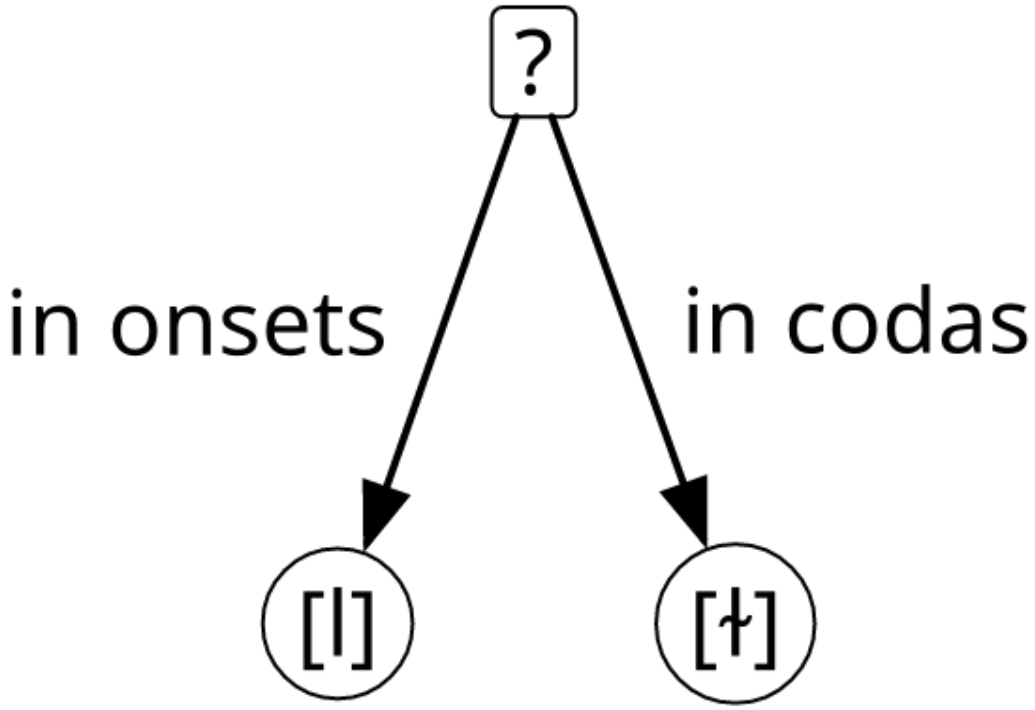
```
tribble(
  ~transcription, ~word,
  !!!str_split("lif leaf
   lup loop
   lɛd lead
   leɪm lame
   bleɪm blame",
   "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
tribble(
  ~transcription, ~word,
  !!!str_split(
    "fiɬ feel
     puɬ pool
     dɛɬ dell
     meɪɬ mail
     hɛɬp help",
   "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
```

| | |
|---|---|
| lif | *leaf* |
| lup | *loop* |
| lɛd | *lead* |
| leɪm | *lame* |
| bleɪm | *blame* |

| | |
|---|---|
| fiɬ | *feel* |
| puɬ | *pool* |
| dɛɬ | *dell* |
| meɪɬ | *mail* |
| hɛɬp | *help* |

## Phonemic Analysis

Neither onsets nor codas are more “basic”




## Neutralization

```
tribble(
  ~transcription, ~word,
  !!!str_split("bɛt bet
pæt pat
ɹəsaɪt recite
bɛd bed
pæd pad
```

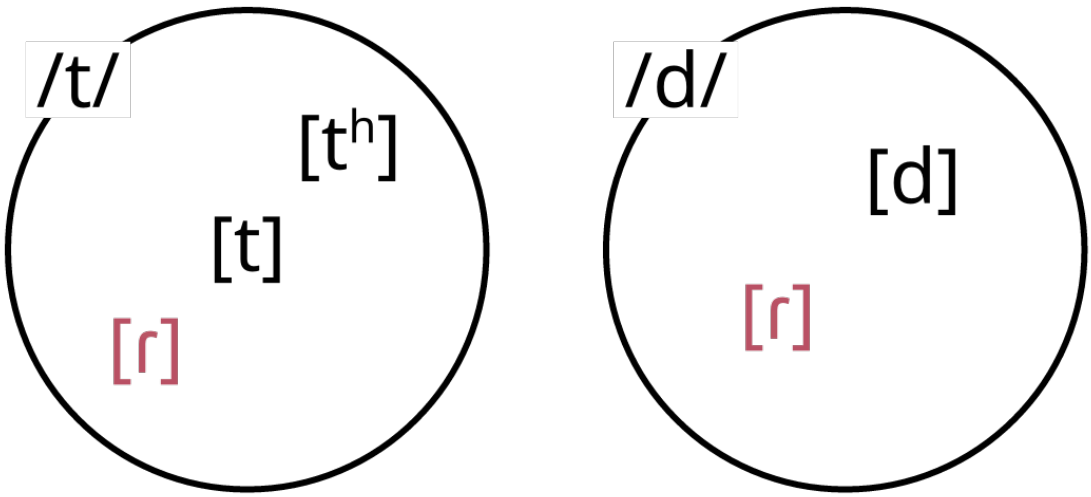


```
bɹaɪd bride",
    "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
tribble(
  ~transcription, ~word,
  !!!str_split(
  "bɛɾɪŋ betting
pæɾɪŋ patting
ɹəsaɪɾ̩ recital
bɛɾɪŋ bedding
pæɾɪŋ padding
bɹaɪɾ̩ bridal",
    "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
```

| | |
|---|---|
| bɛt | *bet* |
| pæt | *pat* |
| ɹəsaɪt | *recite* |
| bɛd | *bed* |
| pæd | *pad* |
| bɹaɪd | *bride* |

| | |
|---|---|
| bɛɾɪŋ | *betting* |
| pæɾɪŋ | *patting* |
| ɹəsaɪɾ̩ | *recital* |
| bɛɾɪŋ | *bedding* |
| pæɾɪŋ | *padding* |
| bɹaɪɾ̩ | *bridal* |

## Neutralization




## Neutralization - Consequences

What evidence could you come up with for the underlying forms of these words?

```
tribble(
  ~transcription, ~word,
  !!!str_split(
  "taɹ̩ tidal
vaɹ̩ vital
taɹ̩ title
spaɹ̩ spider",
   "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
```

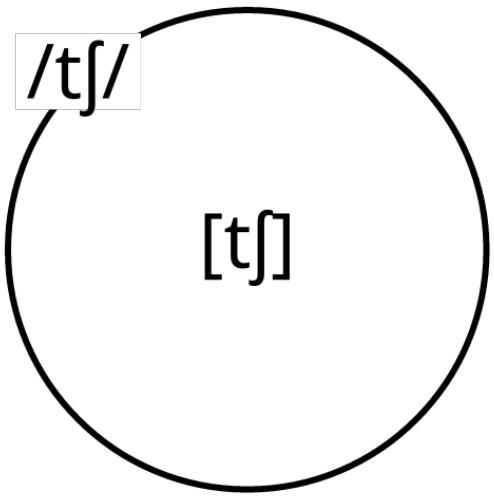
| | |
|---|---|
| taɪɾl̩ | *tidal* |
| vaɪɾl̩ | *vital* |
| taɪɾl̩ | *title* |
| spaɪɾɹ̩ | *spider* |

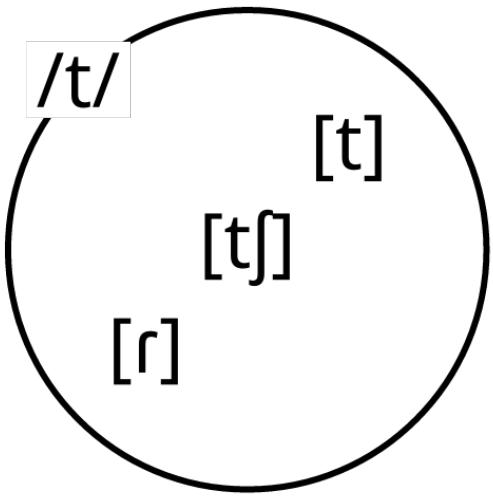
## Neutralization

```
tribble(
  ~transcription, ~word,
  !!!str_split(
  "bɛt bet
bɛtʃju bet-you
bætʃ batch
lɛd lead
lɛdʒju lead-you
lɛdʒ ledge",
   "\\s+"
  )[[1]]
) |>
  mutate(
    word = str_replace(word, pattern = "-", replacement = " ")
  ) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
```

| | |
|---|---|
| bɛt | *bet* |
| bɛtʃju | *bet you* |
| bætʃ | *batch* |
| lɛd | *lead* |
| lɛdʒju | *lead you* |
| lɛdʒ | *ledge* |

## Neutralization

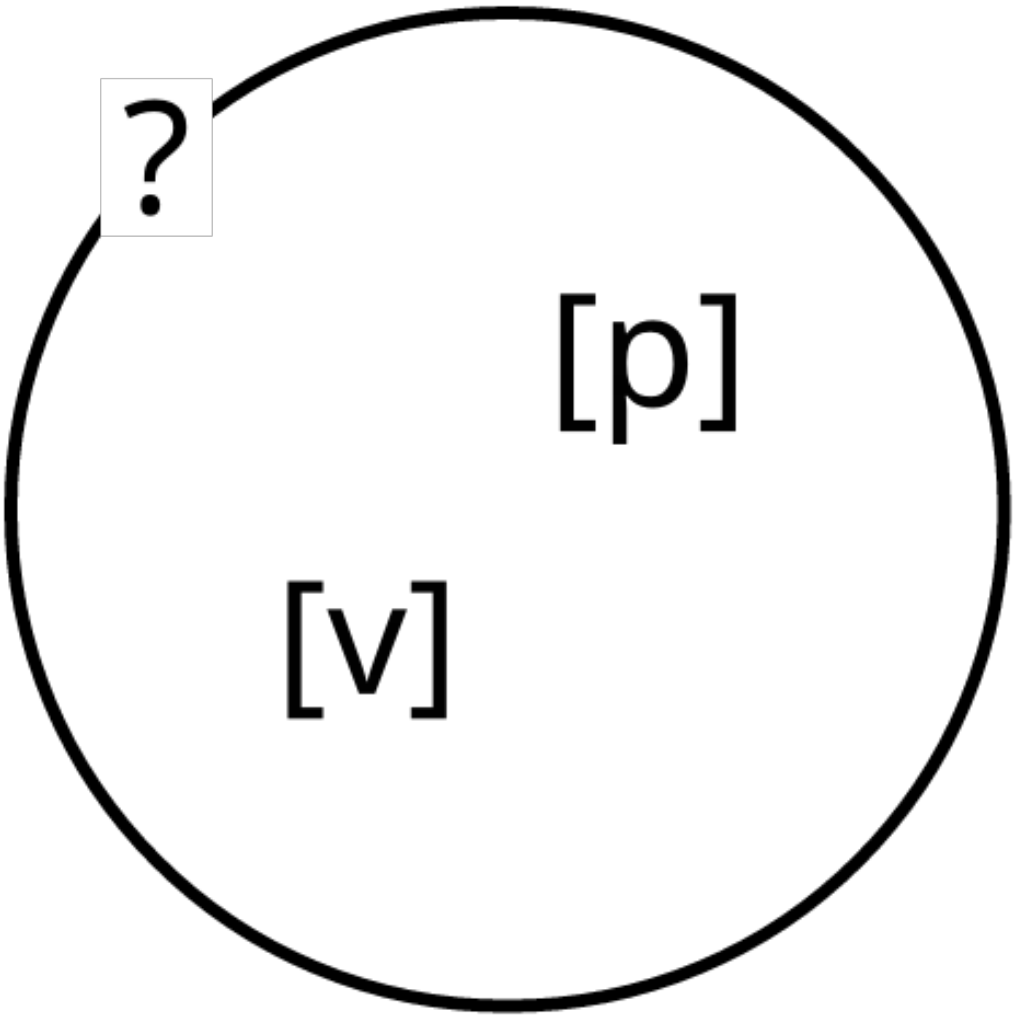



## Abstractness

ʃvovɪç *Schwowisch (a German dialect)*

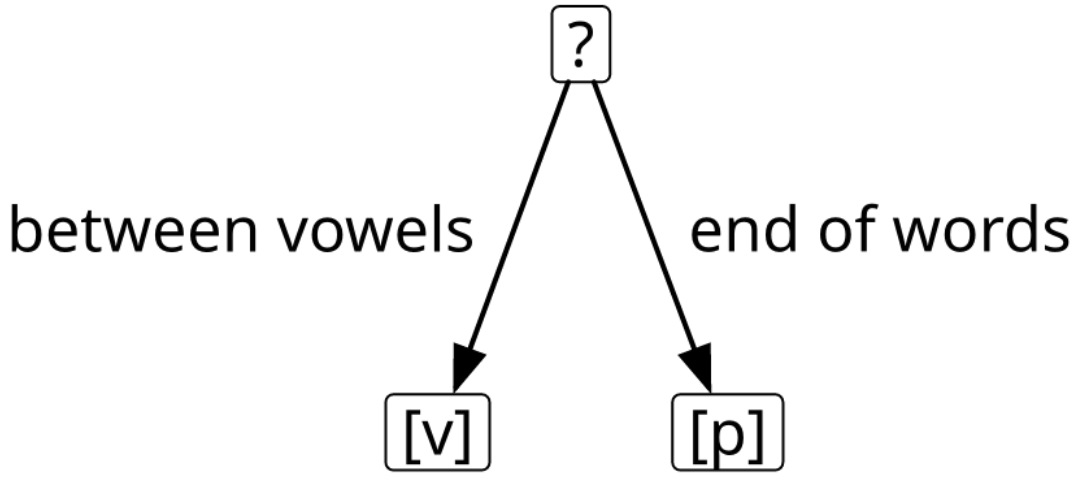
ʃvop *a speaker of Schwowisch*




## Abstractness

ʃvovɪç *Schwowisch (a German dialect)*

ʃvop *a speaker of Schwowisch*




## Abstractness

ʃvovɪç *Schwowisch (a German dialect)*

ʃvop *a speaker of Schwowisch*

